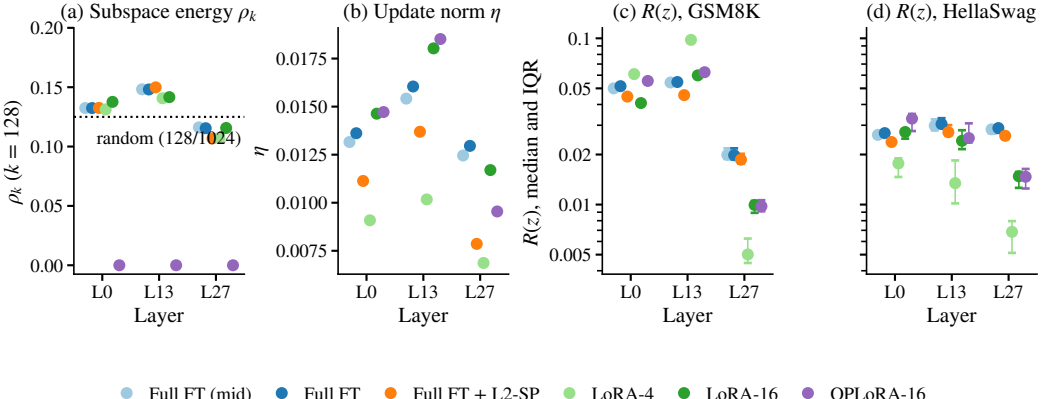
(a) Subspace energy ρ_k
ρ_k (k = 128)
0.20
0.15
0.10
0.05
0.00
random (128/1024)
L0
L13
L27
Layer
(b) Update norm η
η
0.0175
0.0150
0.0125
0.0100
0.0075
L0
L13
L27
Layer
(c) $R(z)$, GSM8K
$R(z)$, median and IQR
0.1
0.05
0.02
0.01
0.005
L0
L13
L27
Layer
(d) $R(z)$, HellaSwag
L0
L13
L27
Layer
Full FT (mid)
Full FT
Full FT + L2-SP
LoRA-4
LoRA-16
OPLoRA-16